

Reads over Atom Spaces: A Unified Mathematical Formulation of Context, Retrieval, and Parameters in Language Models

Farshid Mossaiby¹,

^aDepartment of Civil Engineering, University of Isfahan, 81744-73441, Isfahan, Iran

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¹ **Research Highlights**

- ² • Weighted neural reads are expressed over valued measurable atom spaces.
- ³ • Exact top- k truncation has two bounds and an error-budget rule.
- ⁴ • CounterFact-500 exposes strong retrieval but weak paraphrase recall.
- ⁵ • Signed-read perturbation bounds quantify approximate cancellation failure.
- ⁶ • Five-seed matched routing exposes corpus-to-parameter confusion.

^{*}Corresponding author.